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SOCIAL SCIENCE

Residential segregation by age in the United States

Richelle Winkler^{a*} and Rozalynn Klaas^b

^aDepartment of Social Sciences, Michigan Technological University, 217 Academic Office Building, 1400 Townsend Drive, Houghton, MI 49931, USA; ^bApplied Population Laboratory, University of Wisconsin-Madison, 1450 Linden Drive, Madison, WI 59801, USA

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This map analyzes geographic variation in residential segregation by age in the contiguous United States at Census 2010. We evaluate segregation at the micro-scale between blocks within counties and then examine spatial patterns of segregation by county across the United States using local indicators of spatial autocorrelation. This approach emphasizes local levels of segregation analyzing the extent to which older adults and younger adults reside in the same immediate neighborhood, and it assesses more regional levels of segregation by evaluating spatial clustering. Findings show that the extent of age segregation varies significantly across the contiguous United States. Counties in the Great Plains, in other areas of the Rural West and the Upper Midwest, and in Florida tend to exhibit high segregation, while Appalachia and other areas of the Rural South are generally more integrated.

Keywords: age; segregation; spatial analysis

1. Introduction

The map presented here and its accompanying research project document the extent of residential segregation between older adults and younger adults across the contiguous United States at Census 2010 with particular emphasis on spatial variation in segregation across counties. It makes important contributions to demographic studies of segregation by analyzing age (which has largely been ignored in studies of segregation), by measuring segregation at multiple geographic scales across the entire United States (rather than focusing on metropolitan areas), and by spatially analyzing segregation patterns with maps and exploratory spatial statistics.

The major questions driving this research include:

- To what extent are American neighborhoods segregated by age between older and younger adults?
- How does age segregation vary across geographic space? Are some areas of the country more or less segregated than others?

The population of the United States is aging with multifaceted demographic, social, and economic consequences. One potential consequence is increasing age segregation, but demographers have almost entirely ignored age as a factor in segregation (Cowgill, 1978 is one exception), generally assuming that it is ‘natural’ or benign (Uhlenberg, 2000). However, there are potentially vital implications for understanding political ideologies, securing local service provision, ensuring support for institutions (like schools and senior centers), promoting the health and well-being of the older population who may benefit from interaction with young people, and for transferring intergenerational knowledge (Attias-Donfut, 2000; Binstock, 2010; Foner, 2000). As households, institutions (like school, work, and service centers), and activities (recreation, hobbies, etc.) have become increasingly age-segregated over time; integrated residential communities could provide space where people of different age groups build relationships and share common goals (Riley & Riley, 2000). Without opportunities for interaction, age groups may compete with one another for limited public and private resources to support the interests,

*Corresponding author. Email: rwinkler@mtu.edu



agendas, services, and institutions that best meet their age-specific needs (Binstock, 2010; Foner, 2000; Preston, 1984).

This study focuses on segregation between older adults (age 60 and over) and younger adults (age 20–34) that occurs within the immediate residential neighborhood of the census block. Older adults may have social and political interests in promoting social security, senior care and services, leisure, and aging-related health services. Younger adults, on the other hand, are in prime child-bearing years and may have interests in supporting education, family-oriented policies, and active recreation.

2. Methods

We employ the dissimilarity index (Duncan & Duncan, 1955; Massey & Denton, 1988) to measure residential segregation and to analyze patterns of age segregation at multiple scales – between states, between counties, between county subdivisions (cities, villages, townships and other rural space), and at the micro-scale between blocks within counties. The dissimilarity index (D) is the most commonly used measure of segregation. It measures the evenness of distribution of the population across space, ranging from 0 to 100 where 0 = no segregation and 100 = total segregation. It can be interpreted as the percentage of the population who would have to move to equally distribute older adults and younger adults.

Notably, D is not affected by the composition of the population at the larger geography. For instance, when examining age segregation by block within counties, a county with a high proportion of older population (and a small young population) can have the same D as a county with a low proportion of older population (and large young population). Instead, D is a measure of how evenly dispersed across neighborhoods the population that does reside within the county is.

Data are population counts by five-year age group from US Census 2010. Following Lichter et al. (2007) and Parisi et al. (2011), census blocks provide the basic component building block for our analysis. These units are similar to a city block (in more urban areas and considerably larger in more rural areas). They approximate the immediate neighborhood within which people live and interact with neighbors on a regular basis. While most segregation studies use census tract as unit of analysis, use of the tract has been called into question (Lee et al., 2008), and they are not appropriate for nonmetropolitan areas (Lichter et al., 2007). Census blocks are the most appropriate unit for the building blocks of this study because they approximate the geographic influence of neighboring and because they nest within county subdivisions and counties.

In order to measure segregation at multiple scales, we associated multiple geographic identities (county subdivision, county, and state) with each individual block using ArcGIS and TIGER/Line shapefiles to create a dataset in which blocks are clearly nested within county subdivisions which are located within counties, within states. We then calculated the dissimilarity index (D) at the various scales in Stata using Reardon's "seg" command.

Results show that on average in the United States, there is little age segregation between states, somewhat more between counties and between county subdivisions, and moderate age segregation between census blocks within county subdivisions ($D = 42.6$, meaning that 43% of the population would have to move to a new block in order to achieve perfect integration). At the block level, age segregation is similar to what Lichter et al. (2007) found for segregation between Hispanics and Non-Hispanic Whites ($D = 44.05$), and falls within what they refer to as a "moderate" degree of segregation. Because each of the geographies shown in Table 1 nest within one another, the D values at each scale represent compounding segregation at multiple scales. In other words, there is a small degree of segregation between states, additional segregation between counties within states, more segregation between county subdivisions within counties, and in addition to that is local segregation between blocks within county subdivisions.

These national level statistics offer an indication of the average level of segregation across the United States, but they mask geographic variation in segregation. Table 2 summarizes the dissimilarity index for blocks within

Table 1. Average age segregation (D) at multiple scales, 2010.

Between States/Within US	6.7
Between Counties/Within State	13.0
Between County Subs/Within County	11.6
Between Blocks/Within County Subs (micro)	42.6

Source: Authors' calculations from US Census 2010, Summary File 1.

Table 2. Summary statistics for dissimilarity index within counties.

Mean	45.4
S.D.	9.8
Minimum	13.5
Maximum	87.2

Source: Authors' calculations from US Census 2010, Summary File 1.

counties, showing that county *D* scores range from a low value (highly integrated) of 13.5 in Alaska's Aleutians East Borough which has a diverse population of White's, Native Americans, Asians, and Pacific Islanders and is dominated by the fishing industry to a high value (highly segregated) of 87.2 in Chatahoochee County, Georgia where the young adult population is concentrated on the Fort Benning Army base.

Figure 1 visualizes county variation in *D* across space. Darker blue counties indicate high levels of age segregation, while lighter counties experience relatively low levels of segregation. Counties in the Great Plains, in other areas of the Rural West and Upper Midwest, and in Florida tend to be the most segregated by age. Appalachia and other areas of the Rural South tend to be more integrated.

Finally, we employ spatial statistics to investigate the degree to which counties with similar levels of segregation cluster together in space. Moran's *i* is a global measure of spatial autocorrelation noting the average degree to which like counties cluster together (Anselin, 1988). Overall, counties tend to have similar levels of age segregation as their neighbors, exhibiting a high degree of spatial autocorrelation (Moran's $i = 0.59$, p -value < 0.001 , Queen contiguity spatial weights matrix).

Figure 2 shows where pockets of high and low segregation counties cluster across space using local indicators of spatial autocorrelation (LISA statistics). LISA statistics examine relationships between specific counties and their neighbors. For instance, if a county scores high on the segregation index and so do its neighbors, it is

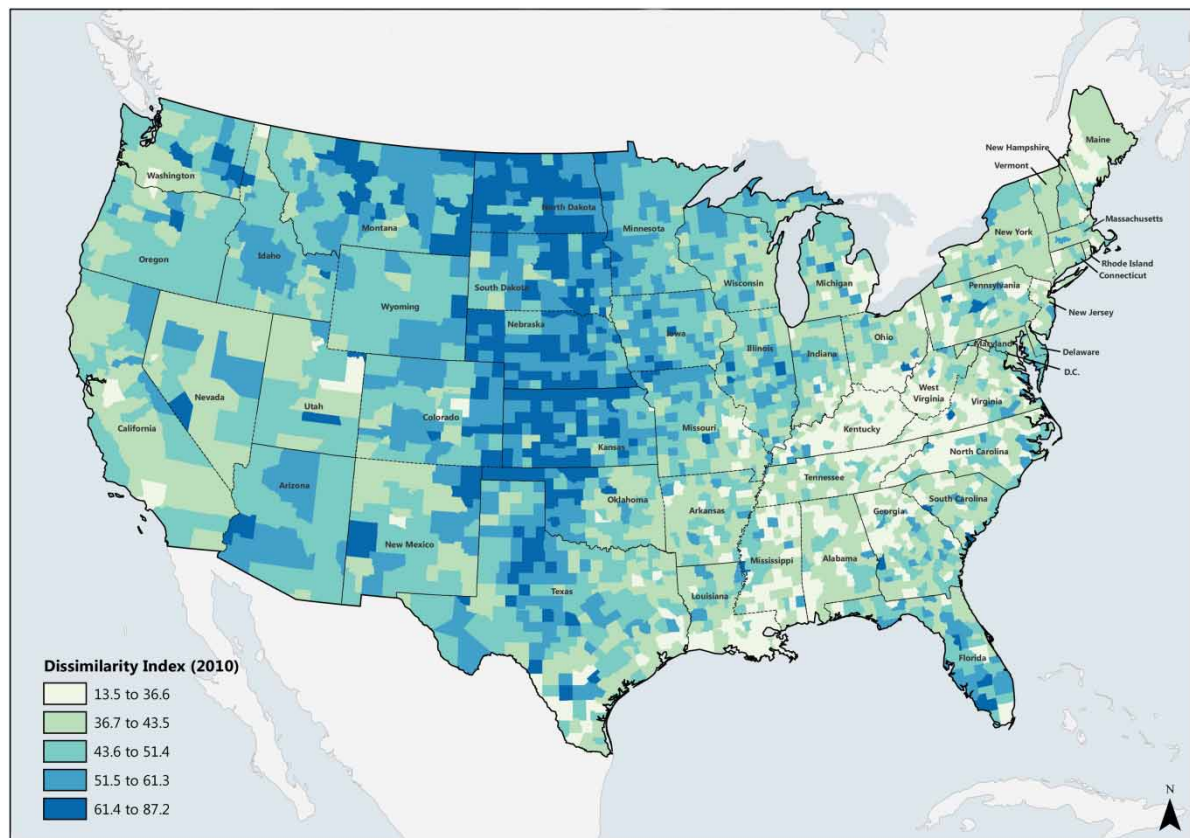


Figure 1. Residential segregation between older and younger adults within counties.

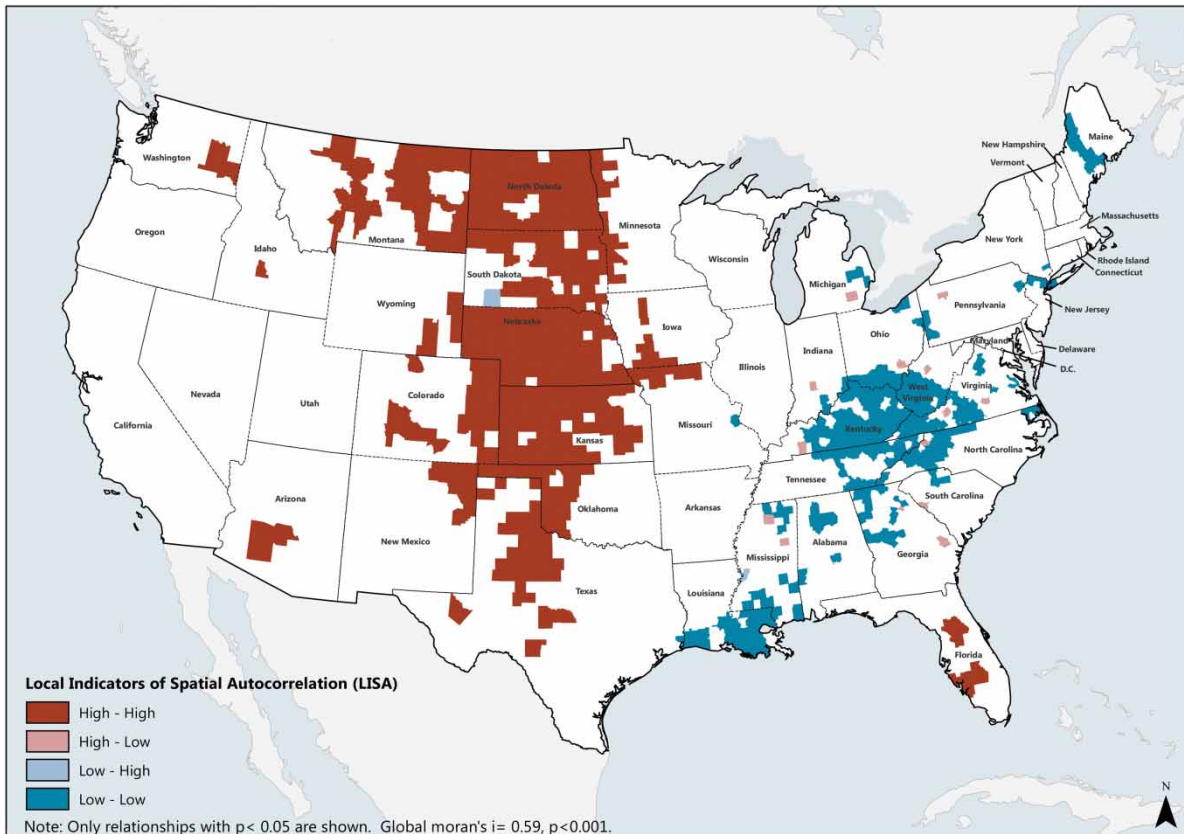


Figure 2. Clusters of age segregation: dissimilarity index of blocks within counties.

shown in dark red. If both a county and its neighbors score low, it is shown in dark blue. Paler shades represent counties that are different from their neighbors, with pink counties exhibiting higher segregation than their neighbors and light blue counties experiencing less segregation than their neighbors. Only counties with statistically significant spatial relationships ($p < 0.05$) are colored. Non-significant counties are shown in white.

3. Conclusions

We find evidence that older adults and younger adults generally live in distinct local neighborhoods, with a similar extent of segregation as Hispanics and non-Hispanic Whites. The degree to which this is true, however, varies considerably across space. While some counties experience significant age segregation, others are quite integrated. Furthermore, counties with similar levels of age segregation tend to cluster together in space creating whole regions of the country with relatively little age integration, especially across the Great Plains and in retirement destination pockets in Florida and Arizona.

Particularly interesting is the fact that these same regions tend to be those with a relatively old population (where older people have moved into the region and/or younger adults moved out). Because the dissimilarity index controls for the age composition of the population at the county level, the finding that certain areas of the country that have an older population composition overall are *also* highly segregated suggests that young adults and older adults may have very little interaction in these areas. While West Virginia, Florida, and North Dakota are each among the oldest states in the country, the relatively few young adults who do live in West Virginia reside in neighborhoods together with the older population to a much greater extent than in North Dakota or Florida.

In sum, this map demonstrates one way in which spatial demographers can contribute to segregation studies by addressing issues of scale. Evaluating segregation at the micro-scale between blocks within counties and then examining spatial patterns of segregation by county across the United States using local indicators of spatial autocorrelation allows for a more comprehensive analysis of the multiple geographic levels in which segregation may

occur. This approach emphasizes local levels of segregation analyzing the extent to which older adults and younger adults reside in the same immediate neighborhood, and it assesses more regional levels of segregation by evaluating county-level patterns and clustering of counties across space. Our substantive results imply that further analysis of age segregation across the United States is warranted and raise questions about the differences in the extent of age segregation between similarly aging areas of the country like the Great Plains and Appalachia.

Software

ArcGIS 9.3 was used to associate block-level data from US Census 2010 on population counts by five-year age group with geographically referenced TIGER/Line shapefiles for census blocks, county subdivisions, counties, and states and to generate dbf files that reference multiple levels of geography for each block. These block data with new fields for associated geography for each of the 50 US states and the District of Columbia were then merged into one large national file using Stata statistical software. Segregation indexes were calculated in Stata and transformed to dbf files using StatTransfer. Indicators of spatial autocorrelation (morán's i and LISA) were generated using the Spatial Analyst extension in ArcGIS 9.3. Maps were generated and stylized in ArcMAP and exported as picture files. The layout of the map page was created with text and pictures incorporated in Adobe InDesign CS5.5 and exported to Adobe pdf.

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